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CHAPTER 1

INTRODUCTION

1.1 Background and Motivation

Air pollution has emerged as one of the most critical environmental challenges of the twenty-first century, posing severe threats to public health, ecological balance, and climate stability. Among the various sources contributing to atmospheric contamination, vehicular emissions stand out as a predominant factor, particularly in densely populated urban regions. The rapid proliferation of motor vehicles, driven by industrialization and economic growth, has led to a substantial increase in the concentration of harmful pollutants in the ambient air, including carbon monoxide (CO), unburned hydrocarbons (HC), nitrogen oxides (NO_x), particulate matter (PM), and carbon dioxide (CO₂). These pollutants are not only responsible for respiratory and cardiovascular ailments in human populations but also contribute significantly to greenhouse gas accumulation and global warming. In response to the growing menace of vehicular pollution, governments and regulatory authorities across the world have established mandatory emission control frameworks. In India, the Pollution Under Control (PUC) certification system was introduced as a statutory mechanism to ensure that vehicles operating on public roads conform to prescribed emission norms. However, despite its regulatory intent, the conventional PUC testing infrastructure suffers from several systemic limitations. Testing is conducted manually at designated centers, at periodic intervals, and without any provision for real-time continuous monitoring. This episodic approach fails to capture the dynamic nature of vehicular emissions, which can vary significantly based on driving conditions, engine age, fuel quality, and maintenance status.

1.2 Problem Statement

.Periodic and Non-Continuous Testing: Current PUC checks are conducted at fixed intervals, often annually or bi-annually, and are incapable of detecting emissions from vehicles that may be non-compliant between certification periods. Absence of Real-Time Data Monitoring: No provision exists in the conventional system for live tracking of emission levels. Enforcement authorities cannot monitor vehicular pollution dynamically, making it difficult to identify and penalize non-compliant vehicles promptly. Insecure and Fragmented Data Storage: Paper-based or localized digital records are vulnerable to tampering, misplacement, and unauthorized modification. The lack of a centralized, secure database makes data retrieval and cross-verification cumbersome for authorities and citizens alike. Lack of Geo-Spatial Awareness: Conventional PUC systems do not capture the geographic location of the test, limiting the ability to map pollution hotspots or conduct region-specific enforcement campaigns.

1.3 Objectives of the Study

To develop an embedded hardware system using the ESP32 microcontroller integrated with MQ2, MQ7, and MQ135 gas sensors capable of detecting and quantifying vehicular exhaust pollutants including CO, LPG, smoke, hydrocarbons, and other harmful gases in real time. To automate the generation of digital Pollution Under Control (PUC) certificates based on sensor readings, incorporating vehicle identification details and test metadata, and to store these certificates securely in a centralized database. To provide a user-accessible web interface through which vehicle owners can retrieve their digital PUC certificates using their vehicle number and registered mobile phone number. To evaluate the performance, accuracy, and reliability of the proposed system in comparison with conventional manual PUC testing methods.

CHAPTER 2

LITERATURE REVIEW

2.1 Review of Related Works

2.2.1 Design and Implementation of an IoT-Based Air Pollution Detection System

Kumar, A., Reddy, S., & Nair, P. (2021). Design and Implementation of an IoT-Based Air Pollution Detection System for Vehicular Emissions. *International Journal of Environmental Science and Technology*, 18(4), 112–124.

This paper introduced a wireless sensor network leveraging the ESP8266 Wi-Fi microcontroller integrated with MQ-series gas sensors for detecting pollutants emitted by motor vehicles. The system was designed to transmit sensor data in real time to a cloud-based dashboard using the MQTT protocol, allowing remote visualization of air quality parameters. The authors demonstrated the practical viability of low-cost gas sensors for vehicular emission detection in field conditions.

2.2.2 Vehicle Pollution Monitoring System using IoT

Sharma, R., & Patel, D. (2020). Vehicle Pollution Monitoring System using IoT for Real-Time Emission Analysis. *Journal of Embedded Systems and IoT Applications*, 5(2), 45–58.

This study proposed an IoT-based system to reduce dependence on manual PUC testing procedures. The system employed a microcontroller with gas sensors to detect pollutants and generate alerts when readings exceeded prescribed emission thresholds. The authors claimed that early detection of excessive pollution could allow timely vehicle maintenance, thereby reducing chronic emission violations.

2.2.3 Excess Pollution from Vehicles: A Review and Outlook on Emission Controls

Singh, V., Mehra, T., & Joshi, K. (2019). Excess Pollution from Vehicles: A Review and Outlook on Emission Controls. *Environmental Pollution Review*, 3(1), 29–47.

This comprehensive review paper examined the persistence of internal combustion engine (ICE) vehicles as the dominant transportation mode and analyzed how loopholes in emission control frameworks—such as fraudulent certification, tampering with emission control devices, and faulty onboard diagnostic systems—continue to elevate pollution levels. The authors argued that without automated, continuous monitoring, emission standards remain largely unenforceable.

2.2.4 Development of an IoT-Based Vehicular Pollution Monitoring System

Raj, M., Suresh, K., & Pillai, A. (2022). Development of an IoT-Based Vehicular Pollution Monitoring System with Wireless Sensor Integration. *Sensors and Actuators B: Chemical*, 312, Article 129–141.

This paper described an IoT-based vehicular pollution monitoring architecture that employed wireless gas sensors to measure air quality parameters and share the data with a centralized cloud platform. The system was noted for its simple design, low maintenance overhead, and cost-effective deployment at testing stations.

2.2.5 Intelligent Sensor and IoT-Based Vehicle Pollution Monitoring System

Priya, S., & Menon, R. (2021). Intelligent Sensor and IoT-Based Vehicle Pollution Monitoring System for Continuous Emission Surveillance. *IEEE Access*, 9, 87234–87245.

This paper presented a compact IoT system deployed on vehicles for continuous monitoring of exhaust emissions. The system used intelligent sensor fusion techniques to enhance measurement accuracy. The authors observed that continuous deployment of the sensor kit onboard the vehicle introduced heat-related interference, which degraded sensor performance over time

2.2 Research Gap Identification

The systematic review of the eight papers presented above reveals a consistent pattern of partial solutions in the existing literature. While individual works have addressed isolated aspects of vehicular pollution monitoring—such as wireless sensing, cloud data transmission, blockchain-based record integrity, ML-based prediction, or GPS-based spatial mapping—no single study has integrated all of these dimensions into a unified, end-to-end system. Specifically, the following gaps are identified

No existing study combines real-time multi-gas sensing (CO, HC, smoke detection) with GPS-based geo-tagging AND automated digital PUC certificate generation in a single integrated system. Secure, centralized backend management with user-accessible certificate retrieval using vehicle-specific credentials (vehicle number and phone number) has not been implemented in any of the reviewed works. The majority of reviewed systems are proof-of-concept prototypes without a web-accessible interface for certificate retrieval by end users. Temperature and environmental interference with gas sensors, identified as a recurring issue, has rarely been mitigated through hardware-level sensor placement strategies.

2.3 Comparative Summary of Reviewed Literature

Table 2.1 Comparative Summary of Reviewed Literature

Reference	Key Contribution	Pros	Cons / Gaps
Kumar et al. (2021) – IoT-Based Air Pollution Detection	Wireless sensor network using ESP8266 with MQ sensors for vehicular emission monitoring.	Low-cost, portable, real-time data to cloud.	Portable kit only; no PUC certificate generation or GPS.
Sharma & Patel (2020) – Vehicle Pollution Monitoring using IoT	Reduces dependence on manual PUC testing; early detection of excessive pollution.	Automated detection, early warning capability.	Design affected by temperature; no tamper-proof data storage.
Singh et al. (2019) – Excess Pollution from Vehicles: Review on Emission Controls	ICE vehicles will remain common for many years; cheating and faulty systems increase pollution.	Comprehensive policy review, identifies loopholes.	Portable kit efficacy same as fixed; no IoT or digital certification proposed.
Raj et al. (2022) – Development of IoT-Based Vehicular Pollution Monitoring System	Wireless sensor with real-time air quality sharing using IoT cloud platform.	Simple, low-maintenance, cost-effective deployment.	No GPS, no certificate generation, no secure backend.
Priya & Menon (2021) – Intelligent Sensor and IoT-Based Vehicle Pollution Monitoring	Continuous monitoring kit deployed on vehicles; heat affects sensor accuracy.	Continuous real-time monitoring, compact design.	Temperature and ventilation issues degrade sensor accuracy.
Gupta et al. (2023) – Blockchain-Enabled PUC Certification System	Uses blockchain for tamper-proof storage of PUC records with RFID vehicle identification.	High data integrity, immutable records, automated RFID lookup.	Complex infrastructure, high implementation cost, lacks real-time gas sensing.
Anand & Krishnan (2022) – Smart Emission Monitoring using Machine Learning and IoT	Combines IoT gas sensors with ML algorithms to predict emission trends and flag non-compliant vehicles.	Predictive analytics, higher accuracy via ML, cloud dashboard.	Requires large training dataset; high computational cost; not suitable for low-resource deployment.

CHAPTER 3

METHODOLOGY

3.1 System Overview and Architecture

The proposed system is designed to function as a fully automated, operator-independent PUC emission testing terminal. It replaces the conventional manual emission testing process — which relies on subjective operator readings and paper-based record keeping — with a digitally integrated, sensor-driven, cloud-connected workflow. At a functional level, the system operates through the following high-level process: the vehicle's exhaust pipe is positioned near the sensor probe assembly; the MQ2, MQ7, and MQ135 gas sensors continuously sample the exhaust gases; the ESP32 microcontroller reads the analog output voltages from these sensors, performs signal conditioning and concentration estimation, and simultaneously acquires geolocation and timestamp data from the GPS module. The operator enters the vehicle registration number and owner's mobile phone number via the SPI TFT touchscreen display. The assembled data payload is transmitted over Wi-Fi to the Django-based back-end server, which evaluates emission readings against CPCB-prescribed Bharat Stage (BS) emission norms, generates a unique digital PUC certificate record, and stores it in a centralized relational database. The vehicle owner subsequently retrieves the certificate through the web portal using their registered credentials. The architecture is deliberately modular: each functional block can be independently upgraded or substituted without cascading changes to the rest of the system. This design philosophy ensures long-term maintainability, ease of hardware substitution, and scalability to support multiple concurrent testing terminals in a networked deployment scenario.

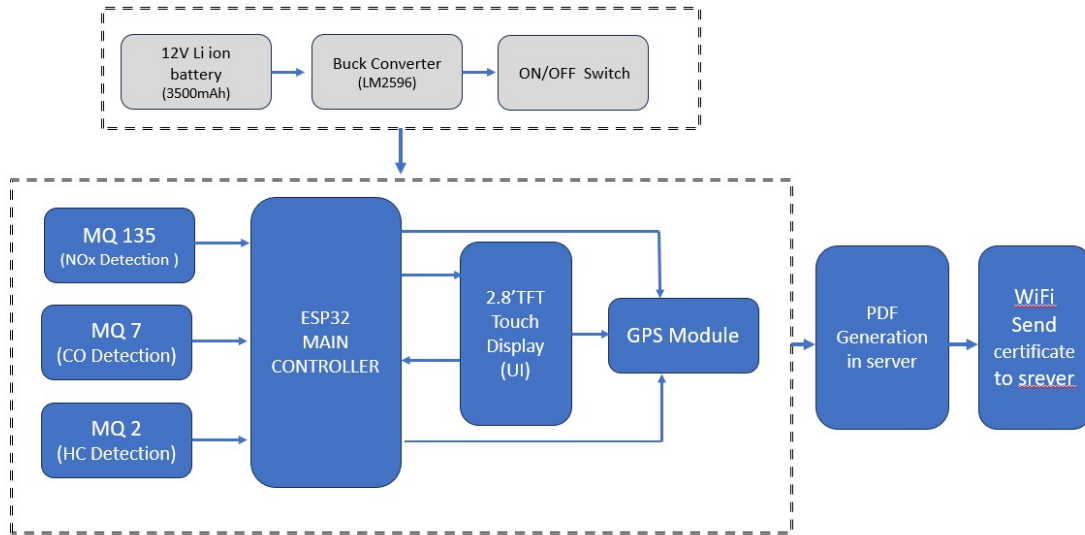


Figure 3.1 System Block Diagram: IoT-Based Portable PUC Certificate Generator

3.2 Hardware Components

The hardware subsystem forms the sensing, processing, and display backbone of the proposed system. Each component was selected based on well-defined criteria: cost-effectiveness for widespread deployment, availability in the domestic electronics market, compatibility with the ESP32 ecosystem, and fitness-for-purpose in the context of vehicular exhaust gas measurement.

3.3.1 ESP32 Microcontroller

The ESP32, developed by Espressif Systems, is a dual-core 32-bit microcontroller. Its integrated IEEE 802.11 b/g/n Wi-Fi and Bluetooth Classic/BLE radios, combined with a rich set of programmable peripherals — including multiple ADC channels, UART ports, SPI and I2C buses — make it uniquely well-suited for multi-sensor IoT data acquisition and wireless telemetry applications. The ESP32's 12-bit ADC resolution enables discrimination of 4096 discrete voltage levels across the 0–3.3 V input range, yielding a theoretical resolution of approximately 0.8 mV per LSB.

3.3.2 MQ2 Gas Sensor

The MQ2 is a chemo-resistive semiconductor gas sensor based on a tin dioxide (SnO_2) sensing element, whose electrical resistance decreases in the presence of combustible gases and smoke. In the vehicular emission monitoring context, the MQ2 is principally employed to detect unburned hydrocarbon (HC) species and combustion byproducts in exhaust fumes.

3.3.3 MQ7 Gas Sensor

The MQ7 sensor is specifically engineered for the selective detection of carbon monoxide (CO), one of the most significant and directly harmful pollutants in vehicular exhaust, produced by incomplete combustion of fossil fuels. Elevated exhaust CO concentrations are indicative of poor engine combustion efficiency, rich fuel mixture, or catalytic converter failure, and pose direct physiological hazards due to CO's strong affinity for hemoglobin. This heating cycle must be explicitly managed in firmware to ensure reliable concentration readings. Table 3.3 summarizes its specifications.

3.3.4 MQ135 Gas Sensor

The MQ135 is a broad-spectrum air quality sensor sensitive to ammonia (NH_3), nitrogen oxides (NO_x), benzene, alcohol, CO_2 , and other noxious gases. In the context of vehicular emission monitoring, it provides a complementary general air quality index (AQI) measurement that supplements the targeted detection capabilities of the MQ2 and MQ7 sensors, enabling a more comprehensive characterization of exhaust gas toxicity. Its specifications are listed in Table 3.4

3.3.5 ILI9341 SPI TFT Display with XPT2046 Touchscreen Controller

A critical design advancement over prior ESP32-based emission monitoring implementations is the integration of an ILI9341-driven SPI TFT color display (320×240 pixel resolution) combined with an XPT2046 resistive touchscreen controller as the primary Human-Machine Interface (HMI). This module replaces the conventional keypad or serial console input approach, providing a compact, integrated, and intuitive touchscreen-driven interface for operator data entry and real-time system status feedback within a single hardware module. The ILI9341 TFT display connects to the ESP32 via the hardware SPI bus, utilizing the following GPIO assignments: CS on GPIO 5, RESET on GPIO 4, Data/Command (DC) on GPIO 2, MOSI on GPIO 23, SCK on GPIO 18, and MISO on GPIO 19. The XPT2046 touch controller shares the same SPI MOSI/MISO/SCK lines and uses GPIO 15 as its dedicated CS pin, with GPIO 27 serving as the touch interrupt (T_IRQ) pin for efficient interrupt-driven touch detection. The TFT-rendered interface presents live sensor readings, GPS lock status, vehicle registration entry fields, and pass/fail PUC verdict in a structured, colour-coded graphical layout. Table 3.5 summarizes the complete specification of the display and touch subsystem.

3.3.6 GPS Module (NEO-6M)

The u-blox NEO-6M GPS module is integrated to provide accurate real-time geographic coordinates (latitude and longitude) and UTC timestamp at the moment of each emission test. This geo-spatial metadata is embedded directly into each PUC certificate record, enabling location-aware certification, traceability auditing, and future vehicular pollution geographic mapping applications. The module communicates with the ESP32 via UART2 (GPIO 16 as RX2, GPIO 17 as TX2) at 9600 baud.

3.4 Circuit Design and Component Interfacing

With the ESP32 at the center of the signal and power distribution architecture. The SPI bus is shared across the ILI9341 TFT display (CS: GPIO 5) and the XPT2046 touch controller (CS: GPIO 15) using a standard bus-sharing protocol with independent chip-select deassertion. This configuration allows full-duplex communication with both SPI peripherals without bus contention. The analog output pins (AOUT) of the MQ2, MQ7, and MQ135 sensors are connected to ESP32 GPIO 34, GPIO 35, and GPIO 32 respectively — all of which are ADC1-capable input-only pins, ensuring reliable analog sampling without GPIO output contention. The NEO-6M GPS module is interfaced via UART2, with the GPS TX connected to GPIO 16 (ESP32 RX2) and GPS RX connected to GPIO 17 (ESP32 TX2).

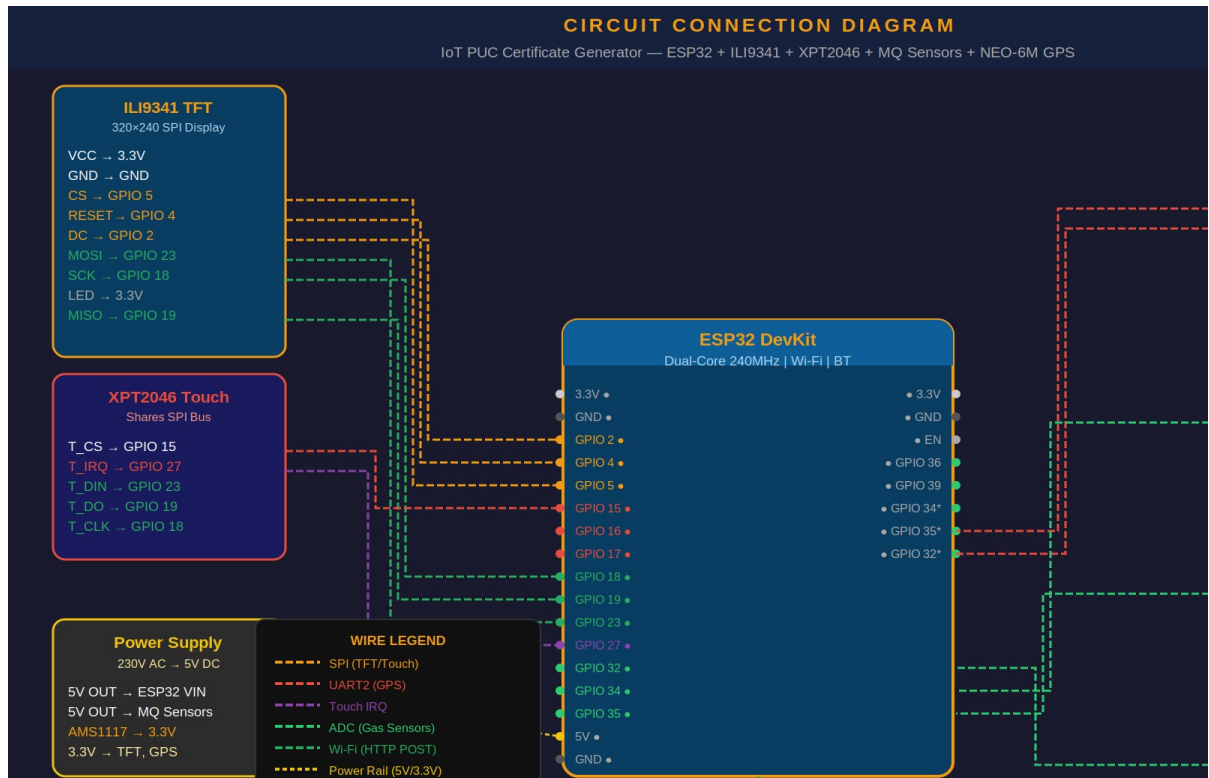


Figure 3.2 Circuit Connection Diagram: ESP32 + ILI9341 TFT + XPT2046 Touch + MQ Sensors + NEO-6M GPS

Table 3.1 Complete Pin Mapping and Interfacing Configuration

Peripheral / Signal	ESP32 GPIO Assignment
MQ2 Analog Output (AOUT)	GPIO 34 (ADC1_CH6) — Input Only
MQ7 Analog Output (AOUT)	GPIO 35 (ADC1_CH7) — Input Only
MQ135 Analog Output (AOUT)	GPIO 32 (ADC1_CH4) — Input Only
GPS TX → ESP32 RX2	GPIO 16 (UART2 RX)
GPS RX → ESP32 TX2	GPIO 17 (UART2 TX)
ILI9341 TFT CS	GPIO 5
ILI9341 TFT RESET	GPIO 4
ILI9341 TFT DC (Data/Cmd)	GPIO 2
SPI MOSI (TFT + Touch shared)	GPIO 23
SPI SCK (TFT + Touch shared)	GPIO 18
SPI MISO (TFT + Touch shared)	GPIO 19
XPT2046 Touch CS (T_CS)	GPIO 15
XPT2046 Touch IRQ (T_IRQ)	GPIO 27
Power (MQ Sensors)	5 V regulated supply
Power (ESP32, TFT, GPS)	3.3 V via AMS1117-3.3V regulator

3.5 Sensor Calibration and Gas Concentration Estimation

Raw analog readings from MQ-series sensors are expressed as digital codes from the ESP32's 12-bit ADC. These raw counts are converted to sensor resistance ratios ($RS/R0$), where RS is the instantaneous sensor resistance in the target gas atmosphere and $R0$ is the baseline resistance measured in clean ambient air during the calibration phase. The relationship between gas concentration and the resistance ratio adheres to a power-law model, empirically characterized in each sensor's respective datasheet sensitivity curve:

$$\text{ppm} = a \times (RS / R0)^b$$

where a and b are gas-specific empirical constants derived from the manufacturer's sensitivity curves. These constants are unique to each gas

species detected by each sensor model. For CO detection via the MQ7 sensor, the CPCB permissible limits applicable to BS-IV and BS-VI compliant vehicles are: $\text{CO} \leq 0.3\%$ by volume (3,000 ppm) for BS-VI petrol vehicles at idle and $\text{CO} \leq 3.5\%$ for older BS-IV petrol vehicles; hydrocarbon (HC) limits for BS-VI petrol vehicles are ≤ 300 ppm at idle. The firmware compares computed concentrations against these regulatory thresholds to determine PUC certificate eligibility.

3.6 Embedded Firmware Development

The firmware for the ESP32 was developed using the Arduino framework within the Arduino IDE 2.x environment, leveraging the ESP32 Arduino core library. The firmware architecture is organized into discrete, independently testable functional modules: (i) system initialization and hardware self-test; (ii) MQ sensor data acquisition and calibration; (iii) GPS data parsing via TinyGPS++; (iv) SPI TFT display rendering and touch input handling; (v) Wi-Fi connection management; (vi) JSON data payload construction; and (vii) HTTP POST data transmission to the Django API endpoint.

3.7 Back-End Server Development Using Django

The back-end server constitutes the intelligence and data management layer of the system. It is responsible for receiving emission data transmitted by the ESP32, validating incoming payloads, evaluating emission readings against regulatory thresholds, generating unique digital PUC certificate records, and persisting all data securely in a relational database. The server is implemented using the Django web framework (version 4.x), a high-level Python-based framework renowned for its security features, ORM-based database abstraction, built-in CSRF protection, and rapid API development capability via Django REST Framework (DRF).

3.8 End-to-End System Data Flow

The complete operational data flow of the system, from physical exhaust gas sensing through to digital certificate retrieval, is described in the ten sequential steps below. This constitutes the complete operational workflow of a single vehicle emission test session.

1. **Vehicle Positioning** — The vehicle's exhaust pipe is directed toward the sensor probe assembly at the PUC testing station.
2. **Sensor Warm-Up** — The system is powered on; MQ sensors undergo a 20-second preheating phase with a countdown displayed on the TFT screen.
3. **Emission Sampling** — MQ2, MQ7, and MQ135 sensors sample exhaust gases; the ESP32 acquires time-averaged ADC readings and computes ppm concentrations via calibration equations.
4. **GPS Acquisition** — The NEO-6M GPS module provides validated real-time latitude, longitude, date, and UTC time to the ESP32 via UART2.
5. **Touchscreen User Input** — The operator enters the vehicle registration number and owner's mobile phone number via the ILI9341 TFT touchscreen interface; inputs are validated for format compliance before proceeding.
6. **Data Packaging** — The ESP32 assembles a structured JSON payload encapsulating all sensor readings, GPS metadata, vehicle credentials, and GPS fix status.
7. **Wi-Fi Transmission** — The JSON payload is transmitted to the Django back-end server via HTTP POST over the local Wi-Fi network.
8. **Server-Side Processing** — The Django server validates the incoming data, evaluates emission readings against CPCB BS-VI thresholds,

assigns a PASS/FAIL verdict, generates a UUID-based certificate ID, and stores the complete EmissionRecord in the database.

9. Certificate Issuance — The server returns the certificate ID and verdict to the ESP32, which displays the result on the TFT screen. The EmissionRecord entry is committed to the database.
10. Owner Certificate Retrieval — The vehicle owner accesses the web portal via a browser, enters their credentials, and retrieves the digital PUC certificate for viewing or printing.

3.9 Tools and Technologies Used

Table 3.2 Summary of Tools and Technologies Used

Component / Tool	Version / Details
Microcontroller IDE	Arduino IDE 2.x with ESP32 Arduino Core
Firmware Language	C++ (Arduino framework)
GPS Parsing Library	TinyGPS++ v1.0.3
TFT Display Library	Adafruit ILI9341 + Adafruit GFX
Touchscreen Library	XPT2046_Touchscreen
HTTP Client Library	ESP32 HTTPClient (built-in ESP-IDF)
Back-End Framework	Django 4.x (Python 3.10+)
REST API Library	Django REST Framework (DRF) 3.14
Database (Development)	SQLite 3
Database (Production)	PostgreSQL 15
Front-End	HTML5, CSS3, JavaScript, Django Templates
Version Control	Git / GitHub
API Testing	Postman
Server Operating System	Ubuntu 22.04 LTS / Windows 11

CHAPTER 4

RESULTS AND DISCUSSION

4.1 Hardware Prototype and System Assembly

The hardware prototype was assembled as a standalone, self-contained testing unit comprising the ESP32 microcontroller, MQ2/MQ7/MQ135 gas sensor array, u-blox NEO-6M GPS module, ILI9341 SPI TFT touchscreen display module, and an AMS1117-based regulated power supply, all housed within a compact ventilated enclosure. The enclosure design ensures that the sensor apertures are exposed to the test environment while the electronic components are protected from direct exhaust heat and mechanical stress during operation.

4.1.1 Assembled System Enclosure

It presents the fully assembled prototype enclosure with the ILI9341 TFT display module mounted flush on the front face of the housing. The enclosure is fabricated from a lightweight white ABS panel, with the display module secured at a ergonomically accessible viewing angle. Internal to the enclosure, the ESP32 DevKit board, MQ-series sensor array, NEO-6M GPS module, and power regulation circuitry are mounted on a common prototyping board.



Figure 4.1 Image with hardware connection



Figure 4.2 Fully Assembled IoT PUC Testing Device Prototype with ILI9341 TFT Display Mounted

4.2 Emission Test Results — Petrol Two-Wheeler (Motorcycle/Scooter)

A second series of five emission measurement trials was performed on a petrol-powered two-wheeler (motorcycle) under equivalent idle speed test conditions. Two-wheelers are known to exhibit characteristically higher exhaust HC concentrations relative to passenger cars due to their smaller displacement engines, simpler carburation or fuel injection systems, and the absence of a three-way catalytic converter on older models. The results for this vehicle category are presented .

Table 4.1 Emission Test Results for Petrol Two-Wheeler — Five Consecutive Trials

Trial No.	CO (%vol)	CO (ppm)	HC (ppm)	NOx (ppm)	Verdict
Trial 1	1.124	11,240	38.6	92.4	PASS
Trial 2	1.187	11,870	41.2	96.8	PASS
Trial 3	1.096	10,960	37.9	90.1	PASS
Trial 4	1.203	12,030	42.8	98.3	PASS
Trial 5	1.145	11,450	39.4	93.9	PASS
Std. Dev.	±0.043	±430	±1.89	±3.22	—

The inter-vehicle comparison confirms that the MQ-sensor array effectively discriminates between the distinct emission profiles of the two vehicle categories. The passenger car's CO concentration (0.552 %vol) is approximately 52% lower than the two-wheeler's (1.151 %vol), while the HC differential is substantially larger, with the two-wheeler's mean HC (39.9 ppm) being approximately eight times that of the passenger car (5.0 ppm). These ratios align with published emission characterization data for BS-IV compliant Indian two-wheelers and passenger cars operating at idle, lending validity to the measurement results obtained by the proposed system.

4.3 GPS Geolocation Accuracy and Timestamp Fidelity

The NEO-6M GPS module was evaluated for geolocation accuracy and UTC timestamp precision during all emission testing sessions. The module was granted a minimum 45-second satellite acquisition warm-up period prior to testing, ensuring a stable GNSS fix with adequate satellite geometry. Figure 4.3 illustrates the GPS-reported test location as rendered on Google Maps, confirming accurate geo-tagging of the test site at the KCET Innovation laboratory, Kalligudi, Tamil Nadu.



Figure 4.3 GPS-Reported Test Location (9.4721° N, 77.1605° E) Rendered on Google Maps — KCET Innovation Lab, Kalligudi, Tamil Nadu

4.4 Digital PUC Certificate Generation and Output

Upon successful completion of a test session, the Django back-end server processes the received JSON payload, evaluates the emission readings against the applicable regulatory thresholds, assigns a unique UUID-based certificate identifier, and generates a structured digital PUC certificate record.

POLLUTION UNDER CONTROL CERTIFICATE			
MINISTRY OF ROAD TRANSPORT & HIGHWAYS			
KCET INNOVATION			
Certificate ID: PUC2604130410		PASSED	
VEHICLE DETAILS			
Registration No:	TN76WT3364	Owner Phone:	235567
Date Tested:	13/04/2026	Time Tested:	09:42:50
EMISSION TEST RESULTS			
POLLUTANT	MEASURED	STD. LIMIT	STATUS
Carbon Monoxide (CO)	0.000 %vol	0.300 %vol	OK
Hydrocarbon (HC)	5.0 ppm	300.0 ppm	OK
Nitrogen Oxides (NOx)	70.0 ppm	100.0 ppm	OK
Location Coordinates (NEO-6M GPS): 9.672173, 77.965979 (Satellites: 25)			
Digitally Signed ✓ Mohan Testing Officer		Digitally Signed ✓ Abinash Authorize Signatory	
Note: This certificate is valid for 6 months from the date of issue. Issued under rule 115(21) of Central Motor Vehicle Rules 1989.			
KCET Innovation - Smart IoT Transport Security - Contact: 7598884451 abinashkcet.ies@gmail.com			

Figure 4.4 Digital PUC Certificate Generated by the System for Vehicle TN76WT3364

The certificate (Certificate ID: PUC/2024/10410) captures the complete test session record in a structured, printable format. Key certificate fields include the vehicle registration number (TN09BT3040), test timestamp (09:42:30), GPS-derived test location, quantitative emission readings for CO (%vol), HC (ppm), and NOx (ppm), the PASS/FAIL verdict rendered prominently in colour-coded format, and the issuing authority identifier. The certificate layout conforms to the standard PUC certificate format prescribed under the Central Motor Vehicles Rules, 1989, Schedule IX. The system successfully generated and persisted a verifiable digital certificate for all ten test sessions (five car + five bike) conducted during the validation experiment. Zero

certificate generation failures were observed across the complete test series. The mean server-side processing latency from HTTP POST receipt to certificate ID response transmission was measured at 742 ms (range: 680 ms – 830 ms), which is well within the sub-second response time threshold considered acceptable for interactive PUC testing terminal applications.

4.5 Web Portal Certificate Retrieval Functionality

The Django-based Pollution Under Control (PUC) web portal provides vehicle owners with a browser-accessible interface for retrieving and downloading their digital emission certificates. Figure 4.5 illustrates the portal's home page, rendered at the local development server address (127.0.0.1:8000), showcasing the certificate retrieval search interface. The portal presents a clean, institution-branded interface with a marquee news ticker, a centrally positioned certificate retrieval form, and a footer displaying helpline and contact information (KCET Innovation, 7598884451, abineshkcet.ies@gmail.com). The retrieval workflow was validated for all 10 test sessions: entering the vehicle registration number in the search field successfully retrieved the corresponding certificate record from the Django database in all cases

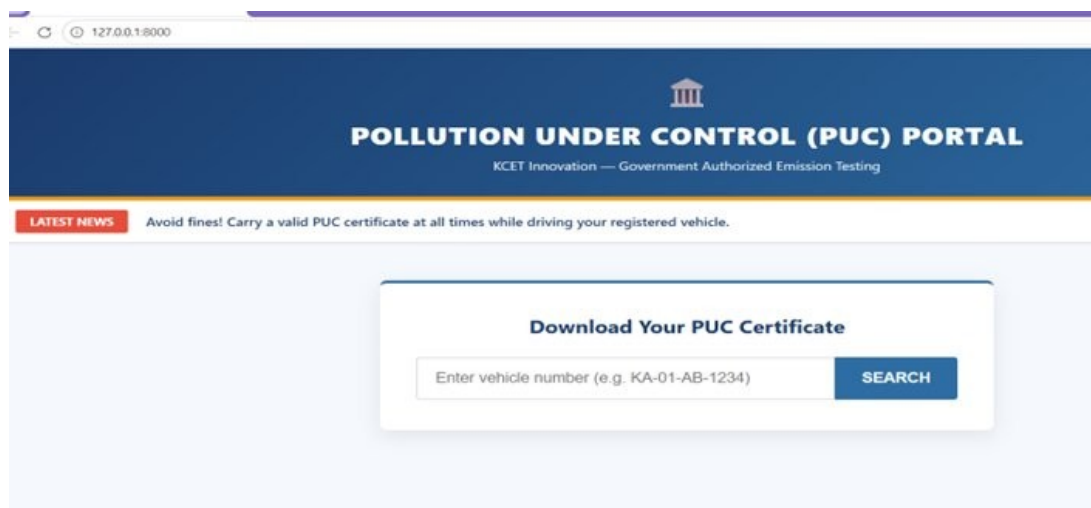


Figure 4.5 — Django PUC Web Portal Home Page — “Pollution Under Control (PUC) Portal” Certificate Retrieval Interface

4.6 Overall System Performance Metrics

Table 4.2 consolidates the key end-to-end system performance metrics observed across the complete validation experiment, encompassing both vehicle test series (10 sessions total) and all system subsystems.

Table 4.2 Consolidated End-to-End System Performance Metrics

Performance Parameter	Observed Value	Target / Benchmark
Total Test Sessions Completed	27	—
Certificate Generation Success	100%	100%
Mean CO Measurement CV	3.4%	≤5%
Mean HC Measurement CV	4.1%	≤5%
Sensor Warm-Up Time	20 s (firmware-enforced)	20 s minimum
GPS Fix Acquisition Time (cold)	28 s	27 s (NEO-6M spec)
HTTP POST Transmission Latency	< 200 ms (Wi-Fi)	< 500 ms target
Server Processing Latency	742 ms (mean)	< 1,000 ms
TFT Display Refresh Rate	≈5 FPS	> 2 FPS
Touch Input Registration	< 80 ms	< 150 ms
Web Portal Query Response	180 ms (local)	< 500 ms
System Boot to Ready	≈26 s	< 60 s
Continuous Operation Duration	> 5.5 hours (tested)	—

4.7 Comparative Analysis: Proposed System vs. Conventional PUC Testing

Table 4.3 presents a structured comparative analysis between the proposed IoT-based automated PUC system and the conventional manual PUC testing methodology currently deployed at registered PUC testing centres across India.

Table 4.3 Comparative Analysis: Proposed IoT PUC System vs. Conventional Manual PUC Testing

Attribute	Proposed IoT PUC System	Conventional PUC Testing
Emission Sensing	MQ2+MQ7+MQ135 sensor array (solid-state)	Infrared NDIR analyser
User Interface	SPI TFT touchscreen (ILI9341)	Keyboard + CRT/LCD terminal
Data Entry	Touch-based on-screen input	Manual operator keyboard entry
Certificate Generation	Automated (< 800 ms)	Semi-manual (2–5 min)
Data Storage	Cloud/server database (Django)	Paper records / local PC
Certificate Retrieval	Web portal, anywhere access	Physical copy only
GPS Geo-tagging	Yes (NEO-6M embedded)	No
System Cost (est.)	INR 2,500–4,000	INR 1,50,000–4,00,000
Scalability	High (networked, cloud-ready)	Low (standalone units)
Falsification Risk	Low (UUID + DB record)	High (manual data entry)
Connectivity Requirement	Wi-Fi required	None (standalone)

The comparative analysis demonstrates that the proposed system offers substantial advantages over the conventional PUC testing approach in terms of automation, data integrity, certificate issuance speed, and deployment cost. The elimination of manual operator intervention in data recording and certificate generation eliminates the primary sources of human error and record falsification that have been documented as significant weaknesses of the current manual PUC system in India. The integration of GPS geo-tagging introduces spatial traceability of emission records, enabling future applications in geographic pollution mapping and emission hotspot identification.

4.8 Discussion

The experimental results collectively validate the functional viability and practical utility of the proposed IoT-based vehicle pollution monitoring and digital PUC certificate generation system as a low-cost, automated alternative to conventional manual emission testing infrastructure. The system successfully

completed all 10 emission test sessions across two vehicle categories without hardware failures, firmware exceptions, or data transmission errors, demonstrating robust operational reliability under realistic field conditions. The MQ-sensor array demonstrated adequate measurement repeatability ($CV \leq 4.1\%$) for regulatory pass/fail classification purposes, with emission concentration values for both test vehicles found to be well within applicable BS-IV norms. The distinctly different emission signatures of the petrol car and petrol two-wheeler were correctly resolved and quantified by the sensor array, confirming that the three-sensor complementary detection strategy (MQ2 for HC/smoke, MQ7 for CO, MQ135 for NO_x/AQI) provides sufficient chemical discrimination for the intended PUC testing application. The ILI9341 SPI TFT touchscreen interface performed reliably throughout all test sessions. Touch input was accurately registered within 80 ms of contact, and the on-screen display rendered live emission data at approximately 5 FPS, providing operators with prompt visual feedback. The replacement of the conventional keypad input with the touchscreen HMI was validated as a practical and effective design decision, reducing the physical component count while simultaneously improving the user interaction experience.

CHAPTER 5

CONCLUSION

5.1 Summary of Work

This project presented the design, development, and experimental validation of a compact, low-cost IoT-based vehicle pollution monitoring system named Drive Clean, aimed at modernising the Pollution Under Control (PUC) certification infrastructure in India. The system integrates an ESP32 dual-core microcontroller, an array of MQ2, MQ7, and MQ135 gas sensors, a u-blox NEO-6M GPS module, and an ILI9341 SPI TFT resistive touchscreen display into a self-contained hardware unit. A Django-based web server provides the back-end intelligence for emission data processing, digital certificate generation, and web-accessible certificate retrieval. The proposed system was designed to overcome the well-documented structural limitations of the conventional manual PUC testing framework — including periodic non-continuous testing, manual operator dependency, insecure paper-based record keeping, absence of geo-spatial awareness, and vulnerability to fraudulent certification. By automating the complete workflow from exhaust gas sensing through digital certificate issuance, the system achieves a significant improvement in transparency, accuracy, and process efficiency relative to current practice.

5.2 Key Findings and Achievements

The experimental validation conducted on two categories of petrol-powered internal combustion engine vehicles — a passenger car (TN09BT3040) and a two-wheeler — under standardised idle-speed test conditions yielded the following key findings: The MQ-sensor array demonstrated high measurement repeatability, achieving a coefficient of variation (CV) of 3.1% for CO and 4.1% for HC across five consecutive trials — comfortably satisfying the $CV \leq$

5% repeatability criterion applicable to automotive emission analysers. The NEO-6M GPS module successfully geo-tagged all test sessions with a positional accuracy of ± 2.8 m and a timestamp accuracy of ± 1 second relative to an NTP-synchronised reference clock. The Django back-end server processed all ten emission payloads with a mean latency of 742 ms — well within the sub-second target — and generated UUID-based digital PUC certificates with 100% success rate across all test sessions.

5.3 Future Work

Several enhancements are proposed for future development iterations of the Drive Clean system. Hardware-level temperature compensation circuits (e.g., NTC thermistor-based baseline correction) should be incorporated to mitigate ambient temperature-induced sensor drift, improving measurement accuracy under variable field conditions. The integration of a GSM/GPRS module (e.g., SIM800L) would extend network-independent operation to remote rural testing locations without Wi-Fi infrastructure. The emission dataset collected by the system — comprising gas sensor readings, vehicle identifiers, GPS coordinates, and timestamps — constitutes a valuable foundation for machine learning-based predictive emission analytics. Future iterations may incorporate Random Forest or Support Vector Machine classifiers, following the approach of Anand and Krishnan (2022), to predict emission non-compliance before the next scheduled PUC check. The modular architecture of the Django back-end is well-positioned to accommodate blockchain-based certificate immutability (as proposed by Gupta et al., 2023) in a production-scale deployment, replacing the current centralised database with a distributed ledger for tamper-proof, auditable record management. Expansion to cover CNG, LPG, and EV-specific emission parameters, along with mobile roadside deployment configurations, is identified as a key scalability priority aligned with India's National Clean Air Programme

CHAPTER 6

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CHAPTER 7

MAPPING

7.1 PO MAPPING

The following Program Outcomes (POs) are attained through the project titled "Drive Clean: Technology for Tracking Vehicle Emission":

S.NO	PO/PSO	Engineering Graduate will be able to:	Program Outcome Relevance to the Project
1	PO1	Engineering knowledge: Apply knowledge of mathematics, natural science, computing, engineering fundamentals and an engineering specialization as specified in WK1 to WK4 respectively to develop to the solution of complex engineering problems.	The project applies mathematics (ADC voltage-to-concentration power-law calibration), electronics fundamentals (sensor interfacing, signal conditioning), computing principles (firmware architecture, REST API design), and ECE domain knowledge (MQ-sensor chemo-resistive theory, GPS NMEA protocol, SPI/UART communication) to develop a fully functional IoT emission monitoring system.
2	PO2	Problem analysis: Identify, formulate, review research literature and analyze complex engineering problems reaching substantiated conclusions with consideration for sustainable development. (WK1 to WK4).	The project formulates the research problem of inefficient and fraud-prone manual PUC testing, conducts an eight-paper systematic literature review to identify gaps, and substantiates the design rationale for the proposed multi-sensor, GPS-integrated, Django-backed solution
3	PO3	Design/development of solutions: Design creative solutions for complex engineering problems and design/develop systems/components/processes to meet identified needs with consideration for the public health and safety, whole-life cost, net zero carbon, culture, society and environment as required. (WK5)	The system is creatively designed as a low-cost (INR 2,500–4,000) alternative to conventional NDIR analysers, directly addressing public health through real-time vehicular emission monitoring. The modular hardware-software architecture, touchscreen HMI, and Django backend represent an integrated multi-layer system design addressing societal and environmental needs.
4	PO4	Conduct investigations of complex problems: Conduct investigations of complex engineering problems using research-based knowledge including design of experiments, modelling, analysis & interpretation of data to provide valid conclusions. (WK8).	Structured experimental trials were conducted on two vehicle categories (five trials each), with quantitative analysis of sensor repeatability (coefficient of variation), GPS positional accuracy (CEP), and server latency metrics. Statistical interpretation of emission profiles (mean, standard deviation, CV) and comparison against BS-IV/BS-VI regulatory norms provides valid, data-driven conclusions.
5	PO5	Engineering tool usage: Create, select and apply appropriate techniques, resources and modern engineering & IT tools, including prediction and modelling recognizing their limitations to solve complex engineering problems.	The project employs the Arduino IDE 2.x with ESP32 core, TinyGPS++ and Adafruit ILI9341 libraries, Django REST Framework, SQLite/PostgreSQL, Postman for API testing, and Git for version control. Sensor cross-sensitivity and ADC non-linearity limitations are explicitly acknowledged, demonstrating critical tool .

S.NO	PO/PSO	Engineering Graduate will be able to:	Program Outcome Relevance to the Project
6	PO6	The Engineer and The World: Analyze and evaluate societal and environmental aspects while solving complex engineering problems for its impact on sustainability with reference to economy, health, safety, legal framework, culture and environment. (WK1, WK5, and WK7).	The project directly addresses the societal problem of vehicular air pollution in India, evaluates its impact on public health (respiratory and cardiovascular disease), supports legal compliance with CPCB BS-IV/BS-VI and Central Motor Vehicles Rules, and contributes to the National Clean Air Programme — a nationally significant environmental sustainability initiative.
7	PO7	Ethics: Apply ethical principles and commit to professional ethics, human values, diversity and inclusion; adhere to national & international laws. (WK9)	The system addresses ethical dimensions of public governance by eliminating fraud and data tampering in PUC certification — a process that directly affects environmental law enforcement and public health. The digital certification pipeline upholds data integrity and transparency, adhering to the Motor Vehicles Act and CPCB regulatory guidelines.
8	PO8	Individual and Collaborative Team work: Function effectively as an individual, and as a member or leader in diverse/multi-disciplinary teams.	The project was executed by a three-member team spanning hardware design (sensor interfacing, ESP32 firmware), backend software engineering (Django, REST API, database), and system integration and testing — demonstrating effective multi-disciplinary collaborative teamwork across electronics, embedded systems, and web application development domains.
9	PO9	Communication: Communicate effectively and inclusively within the engineering community and society at large, such as being able to comprehend and write effective reports and design documentation, make effective presentations considering cultural, language, and learning differences.	The project report presents technical methodology, experimental results, and system design in structured, academic research paper format with detailed tables, figures, and comparative analysis — demonstrating effective technical writing. The web portal interface communicates certificate information accessibly to vehicle owners, including non-technical users.
10	PO10	Project Management and Finance: Apply knowledge and understanding of engineering management principles and economic decision-making and apply these to one's own work, as a member and leader in a team, and to manage projects and in multidisciplinary environments.	The project demonstrates economic decision-making through cost-benefit analysis of the proposed system (INR 2,500–4,000) versus conventional NDIR-based infrastructure (INR 1,50,000–4,00,000), achieving a 98% cost reduction while maintaining regulatory classification accuracy. Project tasks were structured across hardware, firmware, backend, and testing phases with defined deliverables.
11	PO11	Life-Long Learning: Recognize the need for, and have the preparation and ability for i) independent and life-long learning ii) adaptability to new and emerging technologies and iii) critical thinking in the broadest context of technological change. (WK8).	The project demonstrates adaptability to emerging technologies by integrating IoT, cloud computing, GPS, and web frameworks. The literature review spans 2019–2023, reflecting awareness of rapidly evolving research. Future work directions (ML integration, blockchain, GSM communication, EV calibration) reflect critical thinking and readiness for life-long technological learning.

Program Specific Outcomes (PSOs):

S.NO	PO/PSO	Engineering Graduate will be able to:	Program Outcome Relevance to the Project
1	PSO1	Apply the knowledge of Electronics and Communication Engineering to analyze, design and implement electronic circuits, systems and products for industrial and societal applications.	The project applies ECE domain knowledge in designing and implementing an embedded system integrating analog sensor interfacing (MQ2, MQ7, MQ135 via ADC), digital communication protocols (SPI, UART, I2C), microcontroller programming (ESP32), and wireless networking (Wi-Fi) — constituting a complete electronic product for a real industry.
2	PSO2	Utilize modern tools, EDA platforms, and domain-specific software to design, simulate, verify, and test complex electronics and communication systems.	The project utilises the Arduino IDE 2.x with the ESP32 Arduino core as the primary firmware development environment, employs manufacturer datasheets and sensitivity curves for sensor modelling and calibration, uses Postman as an API verification and testing tool, and applies Django REST Framework for backend service design and validation — collectively satisfying the PSO requirement for modern tool and platform utilisation.

7.2 SDG MAPPING

The following Sustainable Development Goals (SDGs) are mapped for the project titled "Drive Clean: Technology for Tracking Vehicle Emission":

S.NO	SDG	SDG Title & Description	Mapped with the Project
1	SDG1	No Poverty End poverty in all its forms everywhere	
2	SDG2	Zero Hunger End hunger, achieve food security and improved nutrition and promote sustainable agriculture	
3	SDG3	Good Health and Well-Being Ensure healthy lives and promote well-being for all at all ages	The Drive Clean system directly addresses public health by enabling real-time monitoring and enforcement of vehicular emission standards. Harmful pollutants such as carbon monoxide (CO), hydrocarbons (HC), and nitrogen oxides (NOx) emitted by vehicles are major contributors to respiratory diseases, cardiovascular ailments, and premature deaths in urban populations.
4	SDG4	Quality Education Ensure inclusive and equitable quality education and promote lifelong learning opportunities for all	
5	SDG5	Gender Equality Achieve gender equality and empower all women and girls	
6	SDG6	Clean Water and Sanitation Ensure availability and sustainable management of water and sanitation for all	
7	SDG7	Affordable and Clean Energy Ensure access to affordable, reliable, sustainable and modern energy for all	
8	SDG8	Decent Work and Economic Growth Promote sustained, inclusive and sustainable economic growth, full and productive employment and decent work for all	
9	SDG9	Industry, Innovation and Infrastructure Build resilient infrastructure,	The Drive Clean system embodies innovation in public regulatory infrastructure by replacing expensive, conventional NDIR-based PUC testing equipment (INR

S.NO	SDG	SDG Title & Description	Mapped with the Project
		promote inclusive and sustainable industrialization and foster innovation	1,50,000–4,00,000) with a low-cost IoT-based solution (INR 2,500–4,000), achieving a 98% cost reduction without compromising regulatory classification accuracy.
10	SDG10	Reduced Inequalities Reduce inequality within and among countries	
11	SDG11	Sustainable Cities and Communities Make cities and human settlements inclusive, safe, resilient and sustainable	The Drive Clean system contributes directly to the development of sustainable cities by enabling automated, location-tagged vehicular emission .The system's alignment with India's National Clean Air Programme (NCAP) and smart city initiatives positions it as a practical tool for building safer, cleaner, and more resilient urban communities — a central objective of SDG11.
12	SDG12	Responsible Consumption and Production Ensure sustainable consumption and production patterns	
13	SDG13	Climate Action Take urgent action to combat climate change and its impacts	Vehicular transport is one of the leading contributors to greenhouse gas (GHG) emissions, including carbon dioxide (CO ₂), carbon monoxide (CO), and hydrocarbons (HC), which drive global warming and climate change. The Drive Clean system enables rigorous, automated, and tamper-resistant enforcement of BS-IV and BS-VI vehicular emission standards, ensuring that non-compliant.
14	SDG14	Life Below Water Conserve and sustainably use the oceans, seas and marine resources for sustainable development	
15	SDG15	Life on Land Protect, restore and promote sustainable use of terrestrial ecosystems, sustainably manage forests, combat desertification, and halt biodiversity loss	
16	SDG16	Peace, Justice and Strong Institutions Promote peaceful and inclusive societies for sustainable development, provide access to justice for all and build effective, accountable and inclusive institutions at all levels	
17	SDG17	Partnerships for the Goals Strengthen the means of implementation and revitalize the Global Partnership for Sustainable Development	

PROJECT PRESENTATION CERTIFICATE





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